

BHV_PeriodicSpeed and BHV_PeriodicSurface

Periodic speed and periodic surface behaviors.

Included MIT MOOS-IvP PDF sources:

- bhv_periodic_speed.pdf
- bhv_periodic_surface.pdf

The PeriodicSpeed Behavior

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1 The PeriodicSpeed Behavior

This behavior will periodically influence the speed of the vehicle while remaining neutral at other times. The timing is specified by a given period in which the influence is on (busy), and a period specifying when the influence is off (lazy), as depicted in Figure 1.

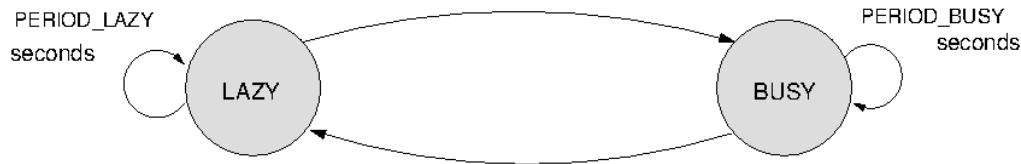


Figure 1: **Busy and Lazy Modes:** In the busy mode the behavior will produce an objective function defined over speed that will potentially influence the speed of the vehicle. In the lazy mode, it simply will not produce an objective function.

It was conceived for use on an AUV equipped with an acoustic modem to periodically slow the vehicle to reduce self-noise and reduce communication difficulty. One can also specify a flag (a MOOS variable and value) to be posted at the start of the period to prompt an outside action such as the start of communication attempts.

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

activeflag:	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
condition:	Specifies a condition that must be met for the behavior to be running. [more] .
duration:	Time in behavior will remain running before declaring completion. [more] .
duration_idle_decay:	When true, duration clock is running even when in the <i>idle</i> state. [more] .
duration_reset:	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
duration_status:	The name of a MOOS variable to which the vehicle duration status is published. [more] .
endflag:	A MOOS variable-value pair posted when the behavior has completed. [more] .
idleflag:	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .
inactiveflag:	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .
name:	The (unique) name of the behavior. [more] .
nostarve:	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
perpetual:	If true allows the behavior to to run even after it has completed. [more] .
post_mapping:	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
priority:	The priority weight of the behavior. [more] .
pwt:	Same as priority .
runflag:	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
spawnflag:	A MOOS variable and a value posted when a behavior is spawned. [more] .
spawnxflag:	A MOOS variable and a value posted when a behavior is spawned. [more] .
templating:	Turns a behavior into a template for spawning behaviors dynamically. [more] .
updates:	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters for the Periodic Speed Behavior.

Parameter	Description
basewidth:	The width of the ZAIC basewidth, in m/sec, in the IvP function.
initially_busy:	If true the initial state is busy. The default is false.
peakwidth:	The width of the ZAIC peakwidth, in m/sec in the IvP function.
period_busy:	The duration of the busy period, in seconds.
period_lazy:	The duration of the lazy period, in seconds.
period_speed:	The desired speed, in m/sec, in the IvP function.

reset_upon_running: Initial conditions reset upon entering the running state. Default is **true**.
summit_delta: The extent of the ZAIC summit delta in the IvP function.

Listing 1.3: Example Configuration Block.

```
Behavior = BHV_PeriodicSpeed
{
  // General Behavior Parameters
  // -----
  name          = transit          // example
  pwt           = 100              // default
  condition     = MODE==TRANSITING // example
  updates       = PSPD_UPDATES     // example

  // Parameters specific to this behavior
  // -----
  basewidth     = 0                // default
  initially_busy = false           // default
  peakwidth     = 0                // default
  period_busy   = 0                // default
  period_lazy   = 0                // default
  period_speed  = 0                // default
  reset_upon_running = true        // default
  summit_delta  = 25               // default
}
```

1.2 Variables Published

Listing 1.4: Variables Published the Periodic Speed Behavior.

Variable	Description
PS_PENDING_BUSY:	Publishes the amount of time in seconds until the behavior reaches the busy mode. During the busy mode this value is zero, and it resets to the value given by the period_lazy parameter upon transitioning into the lazy mode.
PS_PENDING_LAZY:	Publishes the amount of time in seconds until the behavior reaches the lazy mode. During the lazy mode this value is zero, and it resets to the value given by period_busy upon transitioning back into the busy mode. To reduce posting volume, the values posted will be rounded to the nearest second until less than one second remains, after which fractions are posted.
PS_BUSY_COUNT:	This variable is posted by the behavior each time it enters the busy mode. The value is an integer indicating the number of times it has entered the busy mode, posting zero initially.

1.3 State Transition Policy and Initial Condition Parameters

The behavior alternates between one of two modes, the busy mode or the lazy mode. In the former, it will produce an objective function over the *speed* decision variable, and in the latter mode it will simply refrain from producing the objective function. Note that these modes are different from the general behavior run states described in Section ???. If this behavior is idle, i.e., has not met the

conditions for being in the running state, it will not generate an objective function regardless of whether it is in the busy or lazy mode.

When the behavior enters the busy mode, it resets a timer which counts down from `period_busy` seconds, after which it enters the lazy mode. When it enters the lazy mode, it resets a timer which counts down from `period_lazy` seconds, after which it goes back to the busy mode. By default the behavior is initially in the lazy mode, but it can be configured in the opposite manner by setting `initially_busy` to `true`.

By default, when `reset_upon_running` is true, each time the behavior enters the running state, the busy/lazy mode is set to its initial value, and all timers are reset to their initial values. This is true regardless of whether it is entering the running state upon initial helm engagement, or due to transitioning from the idle state. This can be changed by setting `reset_upon_running` to false. In this case the timers are counting down immediately upon helm engagement and continue to do so regardless of the behavior run state.

The PeriodicSurface Behavior

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1 The PeriodicSurface Behavior

This behavior will periodically influence the depth and speed of the vehicle while remaining neutral at other times. The purpose is to bring the vehicle to the surface periodically to achieve some event specified by the user, typically the receipt of a GPS fix. Once this event is achieved, the behavior resets its internal clock to a given period length and will remain idle until a clock time-out occurs.

1.1 Configuration Parameters

Listing 1.1: Configuration Parameters Common to All Behaviors.

<code>activeflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>active</i> state. [more] .
<code>condition</code> :	Specifies a condition that must be met for the behavior to be running. [more] .
<code>duration</code> :	Time in behavior will remain running before declaring completion. [more] .
<code>duration_idle_decay</code> :	When true, duration clock is running even when in the <i>idle</i> state. [more] .
<code>duration_reset</code> :	A variable-pair such as <code>MY_RESET=true</code> , that will trigger a duration reset. [more] .
<code>duration_status</code> :	The name of a MOOS variable to which the vehicle duration status is published. [more] .
<code>endflag</code> :	A MOOS variable-value pair posted when the behavior has completed. [more] .
<code>idleflag</code> :	A MOOS variable-value pair posted when the behavior is in the <i>idle</i> state. [more] .
<code>inactiveflag</code> :	A MOOS variable-value posted when the behavior is <i>not</i> in the <i>active</i> state. [more] .

name:	The (unique) name of the behavior. [more] .
nostarve:	Allows a behavior to assert a maximum staleness for a MOOS variable. [more] .
perpetual:	If true allows the behavior to run even after it has completed. [more] .
post_mapping:	Re-direct behavior output normally to one MOOS variable to another instead. [more] .
priority:	The priority weight of the behavior. [more] .
pwt:	Same as priority .
runflag:	A MOOS variable and a value posted when a behavior has met its conditions. [more] .
spawnflag:	A MOOS variable and a value posted when a behavior is spawned. [more] .
spawnxflag:	A MOOS variable and a value posted when a behavior is spawned. [more] .
templating:	Turns a behavior into a template for spawning behaviors dynamically. [more] .
updates:	A MOOS variable from which behavior parameter updates are read dynamically. [more] .

Listing 1.2: Configuration Parameters for the PeriodicSurface Behavior.

Parameter	Description
acomms_mark_variable	The incoming MOOS variable for resetting the acomms period clock.
ascent_grade	Manner in which desired speed approaches zero on the approach to the surface. The default is linear . Section 1.2.6 .
ascent_speed	Desired speed of the vehicle during the ascent mode. Section 1.2.5 .
atsurface_status_var	MOOS variable indicating number of seconds at the surface. Section 1.2.4 .
mark_variable	The incoming MOOS variable for resetting the period clock. Section 1.2.2 .
max_time_at_surface	The maximum time, in seconds, the vehicle will wait at the surface. Section 1.2.8 .
pending_status_var	MOOS variable written to with remaining time on idle clock. Section 1.2.3 .
period	Duration of the WAITING mode. Section 1.2.1 .
zero_speed_depth	The depth, in meters, at which the desired speed becomes zero on ascent. Section 1.2.7 .

Listing 1.3: Example Configuration Block.

```

Behavior = BHV_PeriodicSurface
{
  // General Behavior Parameters
  // -----
  name          = periodic_surface    // example
  pwt           = 100                  // default
  condition     = MODE==TRANSITING    // example
  updates       = PSURFACE_UPDATES    // example

  // Parameters specific to this behavior
  // -----
  acommms_mark_variable = ACOMMS_RECEIVED // example
  ascent_grade = linear                  // default
  ascent_speed = -1                      // default
  atsurface_status_var = TIME_AT_SURFACE // default
  mark_variable = GPS_UPDATE_RECEIVED    // default
  max_time_at_surface = 300              // default
  pending_status_var = PENDING_SURFACE   // default
  period = TIME_AT_SURFACE               // default
  zero_speed_depth = 0                   // default
}

```

1.2 Detailed Discussions of Behavior Parameters

1.2.1 The `period` Parameter

The `period` parameter sets the duration of the period, in seconds, during which the behavior will remain in the `IDLE_WAITING` state.

1.2.2 The `mark_variable` Parameter

The `mark_variable` names a variable used for indicating when the behavior witnesses the event that would reset the period clock. On each iteration, the variable is checked against its last known value and if different, the clock is reset. The default value for this parameter is `GPS_UPDATE_RECEIVED`. If this variable is populated by another process with a value indicating the time a GPS fix is obtained, then the mark will occur on each GPS fix. Since the value of this argument names a MOOS variable, it is case sensitive.

1.2.3 The `pending_status_var` Parameter

The `pending_status_var` names a variable to be written to with the value of the remaining time on the idle clock, rounded to integer seconds. The default value is `PENDING_SURFACE`. Since the value of this argument names a MOOS variable, it is case sensitive.

1.2.4 The `atsurface_status_var` Parameter

The `atsurface_status_var` parameter names a variable to be written to with the number of seconds that the vehicle has been waiting at the surface (for the event indicated by the MOOS variable specified in the `mark_variable` parameter). The number of seconds is rounded to the nearest integer

and will be zero when the vehicle is not at the surface. The default value is `time.at.surface`. Since the value of this parameter names a MOOS variable, it is case sensitive.

1.2.5 The `ascent_speed` Parameter

The `ascent_speed` parameter indicates the desired speed (m/s) of the vehicle during the ascent state. If left unspecified, the ascent speed will be equal to the current noted speed at moment it transitions into the ascent state.

1.2.6 The `ascent_grade` Parameter

The `ascent_grade` parameter indicates the manner in which the ascent speed approaches zero as the vehicle progresses toward the `zero_speed_depth`. It has four legal values: `fullspeed`, `linear`, `quadratic`, and `quasi`. The default is `linear`. In all four cases, the initial speed is determined by the parameter `ascent_speed`, and the desired speed will be zero once the `zero_speed_depth` has been achieved. The four settings determine the manner of slowing to zero speed during the ascent. The `fullspeed` setting indicates that desired speed should remain constant through the ascent right up to the instant the vehicle achieves `zero_speed_depth`. For the other three settings the speed reduction is relative to the starting depth (the depth noted at the outset of the ascent state) and the `zero_speed_depth`. With the `linear` setting, the speed reduction is linear. With the `quadratic` setting, the speed reduction is quadratic (quicker initial speed reduction). With the `quasi` setting the speed reduction is between linear and quadratic. The value passed to this parameter is not case sensitive.

1.2.7 The `zero_speed_depth` Parameter

The `zero_speed_depth` parameter sets the depth (in meters) during the ascent state at which the desired speed becomes zero, and presumably further ascent is achieved through positive buoyancy.

1.2.8 The `max_time_at_surface` Parameter

The `max_time_at_surface` parameter sets the maximum time (in seconds) spent in the `AT_SURFACE` state, waiting for the event indicated by the `mvar.variable`, before the behavior transitions into the `IDLE` state.

1.3 Internal States of Periodic Surface Behavior

The behavior can be in one of four states as described in Figure 1 below.

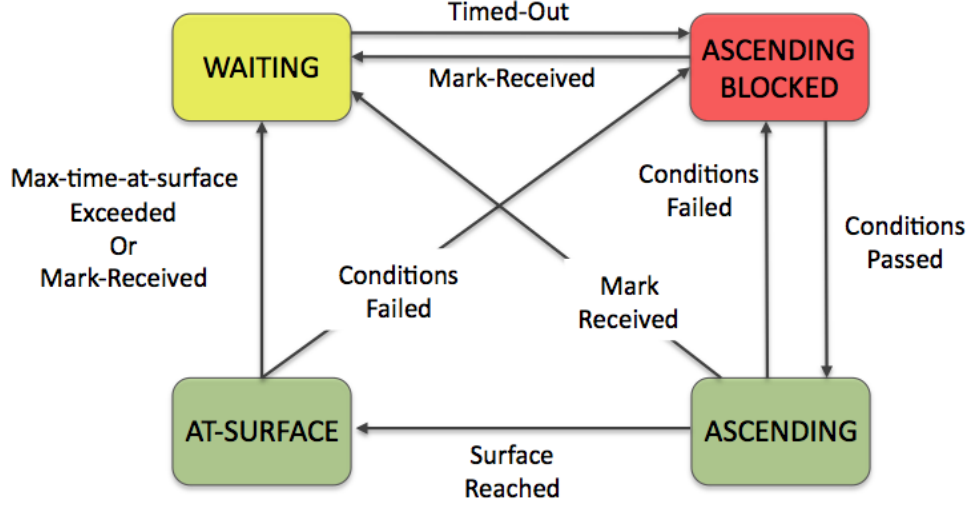


Figure 1: Possible modes of the PeriodicSurface behavior.

In the **WAITING** state the behavior is simply waiting for its clock to wind down to zero. The duration is given by the **period** parameter listed below. The clock is active despite any other run conditions that may apply to the behavior. It is started when the behavior is first instantiated and also when the desired event occurs at the surface. The **ASCENDING_BLOCKED** state indicates that the behavior timer has reached zero, but another run condition has not been met. This is to prevent the behavior from trying to surface the vehicle when other circumstances override the need to surface. In the **ASCENDING** state, the behavior will produce an objective function over depth and speed to bring the vehicle to the surface. A couple parameters described below can determine the trajectory of the vehicle during ascent. This state can transition back to the **ASCENDING_BLOCKED** state if run conditions become no longer satisfied prior to the vehicle reaching the surface. In the **AT_SURFACE** state the vehicle is at the surface waiting for a specified event.